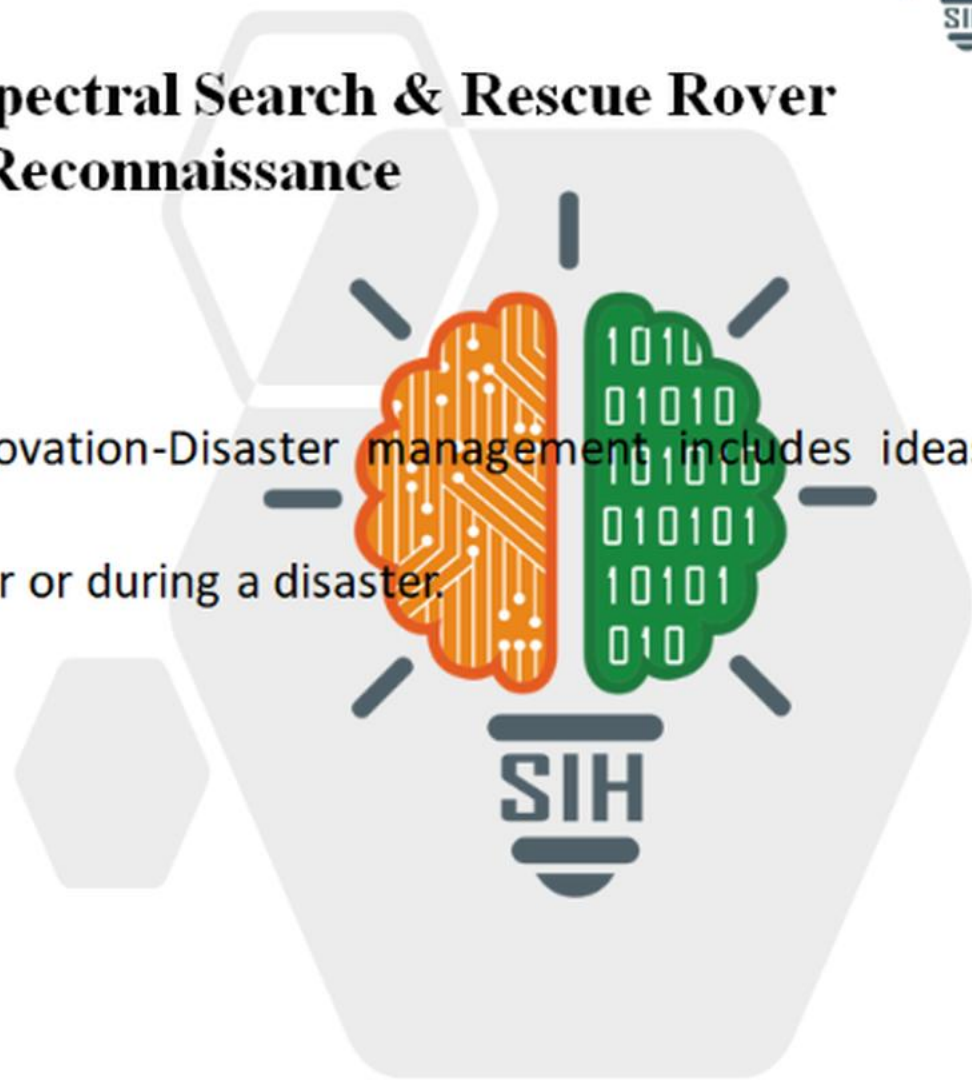




**ARIES — Advanced Multi-Spectral Search & Rescue Rover
for Disaster Reconnaissance**

- **Problem Statement ID – 26223**
- **Problem Statement Title-** Student Innovation-Disaster management includes ideas related to risk mitigation, Planning and management before, after or during a disaster.
- **Theme-** Disaster Management
- **Team I'D-** SIH_1186
- **PS Category-** Hardware
- **Team Name-** ALPHA-X



10x-20x Cost Disruption

₹50,000 vs ₹25L imported

5 km Deep Rubble LoRa

Sub-GHz concrete mesh

1000°C Fire Armor Shield

Silica Aerogel insulation

3D Blueprint Tactical NDRF Map

Triage severity & safe path

AUTONOMOUS OPERATIONAL PATHWAY (HOW ARIES EXECUTES RESCUE):

01

Autonomous Rubble Ingress & SLAM

Deployed to disaster zone -> Rover navigates ruins autonomously; LiDAR & sonar map safe ingress path.

02

Auto-Trench Air-Hop & Flood Float

If deep chasm/trench met -> 8-rotor octocopter lifts chassis across; pontoon hull floats over flood.

03

Bionic Arm Crevice Probe & Vitals Exam

Spots heat signature -> 4-DOF arm enters 10cm crevice to scan survivor pulse & SpO2 at tip.

04

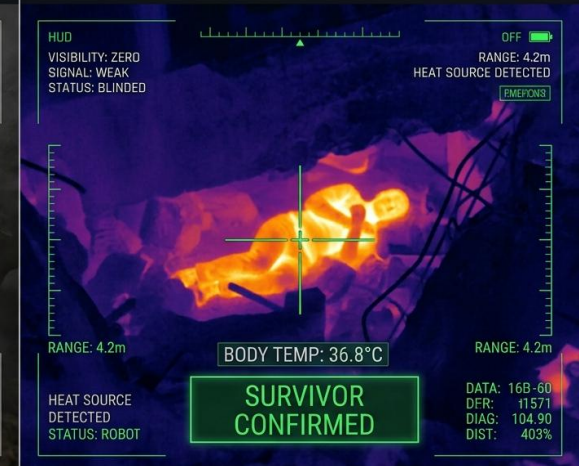
Instant LoRa & GSM Telemetry to NDRF

Transmits physiological vitals + toxic gas report instantly via 5km Sub-GHz LoRa mesh to command.

05

NDRF 3D Tactical Rescue Blueprint

Generates 3D Map for NDRF: exact victim locations, triage injury priority, and safe rescuer corridor!

Standard Optical Camera**ARIES FLIR Thermal Vision**

Comparison of Robot Camera Systems in High-Smoke, Zero-Visibility Environments - [ARIES Project 2024]

THE NDRF TACTICAL EXTRACTION BLUEPRINT (RESCUER SAFETY):

- Pinpointed Locations: Exact 3D coordinates & depth of every trapped survivor.
- Smart Triage Severity: ● Critical (<90% SpO2 extract 1st) vs ● Urgent vs ● Stable.
- Safe Rescuer Corridor: Safe path avoiding secondary wall collapse & chasms.
- Hazard Overlay: Flags MQ-135 toxic gas pockets & 500°C fire zones before entry.

Mission Value: Complete autonomous loop from ingress to survivor vitals triage and final NDRF Tactical Blueprint generation.

Tri-Phibian Mobility

- Land: Kevlar caterpillar tank tracks with high-torque DC planetary gearmotors (climbs 45° debris).
- Water: IP68 sealed carbon boat hull + twin submersible waterjets.
- Air: 8x heavy-lift carbon-fiber outrigger rotors (lifts full 15kg chassis over trenches).

Perception & Bionic Arm

- Bionic Arm: 4-DOF carbon manipulator (1m reach inside 10cm crevice voids).
- Biosensor Tip: MAX30102 pulse oximeter (BPM & SpO2) + BMP280 depth probe.
- Vision Core: FLIR radiometric LWIR thermal core + 3D LiDAR SLAM.

Dual-MCU & Fire Shield

- Dual-MCU: Arduino #1 (Drive) + Arduino #2 (Sensors) - Zero EMF freeze!
- Comms: 5 km Sub-GHz LoRa Mesh (868MHz) + Wi-Fi 6 + SIM800L GSM.
- Fire Armor: 10mm Silica Aerogel (blocks 1000°C) + Nitrogen cold purge.

SYSTEM ARCHITECTURE & END-TO-END DATAFLOW PIPELINE:

1. SENSING & ARM

MQ-135, FLIR, SpO2, Sonar

2. DUAL-MCU CORE

Arduino #1 & #2 Isolated Bus

3. RESILIENT COMMS

5 km LoRa Mesh + GSM SMS

4. NDRF TACTICAL HUB

3D Blueprint & Triage Console

Zero-Lockup Guarantee: Dedicated Dual-MCU domain isolation prevents high-current motor EMF noise from disrupting mission-critical gas & vital sensors.

⚠️ **CHALLENGE: Deep Rubble Signal Loss**
2.4GHz Wi-Fi drops through 2m concrete.



🛡️ **ARIES ENGINEERING MITIGATION:**
5 km Sub-GHz LoRa Mesh (868MHz) + SIM800L GSM SMS fallback.

⚠️ **CHALLENGE: Cliffs & Trench Trap**
Rover gets trapped in sudden drop-offs.



🛡️ **ARIES ENGINEERING MITIGATION:**
Integrated 8-Rotor Octocopter automatically lifts chassis across.

⚠️ **CHALLENGE: 500°C Industrial Fire**
Extreme heat fries sensitive microcontrollers.



🛡️ **ARIES ENGINEERING MITIGATION:**
Silica Aerogel Heat Shield (1000°C block) + Nitrogen cold purge.

⚠️ **CHALLENGE: False Alarms & Blind Entry**
Embers/pipes flag false human heat; rescuers blind.



🛡️ **ARIES ENGINEERING MITIGATION:**
MAX30102 SpO2 & pulse check + 3D Tactical Extraction Blueprint.

- 💰 **ECONOMIC VIABILITY & 10x COST ADVANTAGE:**
- Commercial target under ₹50,000 INR vs ₹25-40 Lakhs imported military robots.
 - Built on modular COTS hardware, eliminating expensive overseas maintenance.

- 🔬 **WORKING PROOF OF CONCEPT (PoC) VALIDATION:**
- Scaled prototype validated on dual-microcontrollers with L293D shield.
 - Tested with live gas telemetry, sonar auto-stop, OLED display & Web Serial Hub.

Zero False-Alarm Architecture: Two-tier validation combines physiological SpO2 pulse verification with Incident Commander HUD confirmation.

♥ Social & Life-Saving

- Zero First-Responder Loss: Eliminates sniffer dog & rescuer hazards.
- Golden Hour Acceleration: Spots trapped survivors in minutes via thermal vision.
- Pre-Staged Medical Triage: Medics receive SpO2 & pulse before physical entry.

💰 Economic Democratization

- 10x-20x Cost Reduction: Under ₹50,000 INR vs ₹25-40 Lakhs imported robots.
- COTS Modular Maintenance: Standard parts, zero foreign service lock-in.
- Scalable Pan-India Outreach: Affordable for every municipal fire station.

🌱 Environmental & Mine Safety

- Real-Time Hazard Containment: MQ-135 tracks gas diffusion before civilian impact.
- Zero-Emission Electric Locomotion: Safe in combustible & low-oxygen shafts.
- Deep Underground Recon: Safe for subterranean tunnels and coal mines.



NDRF & State SDRF

Earthquake & building collapses

Fire & Emergency

Basement fires & chemical leaks

Mining Authorities

Deep coal mine & tunnel collapse

Industrial Plants

Hazardous industrial plant scouts

Humanitarian Impact: Aligned with National Disaster Management Plan (NDMP) benchmarks to save lives during critical Golden Hour extractions.

Bill of Materials & Cost Disruption

- Basic Working Prototype (Table Model): Under ₹8,000 INR.
- ARIES 2.0 Enterprise Tri-Phibian Model: Under ₹50,000 INR.
- Imported Military Competitors: ₹25,00,000 to ₹40,00,000 INR.
- Result: 10x to 40x Unprecedented Cost Democratization!

National Guidelines (Govt. of India)

- NDMA Guidelines on Urban Search & Rescue (USAR) Equipment Standards.
- NDRF Standard Operating Procedures (SOP) for Confined Space Hazards.
- Aligned with National Disaster Management Plan (NDMP 2019).
- Hazardous gas threshold compliance (MQ/PID Standards).

Academic USAR Robotics Literature

- Murphy, R. R. (2014) Disaster Robotics, MIT Press / IEEE Transactions.
- Kumar & Sharma (2022) IoT USAR Ground Vehicles, IEEE Xplore / ICRA.
- IEEE IoT Journal: Sub-GHz Wireless Mesh for Deep-Rubble Disaster Telemetry.
- Journal of Field Robotics: Multi-Spectral Thermal Fusion in Fire Zones.

Open-Source Repositories & Live Hubs

- GitHub Repo: github.com/harshitsharma120109-web/ARIES-Search-Rescue-Rover
- Live Web Mission Hub: melodic-halva-b0fa56.netlify.app
- Full Dual-MCU Firmware, Schematics & Web Serial telemetry open-sourced.
- Validated on functional scaled embedded prototype.

OFFICIAL PROJECT REPOSITORY & LIVE WEB MISSION HUB:

- GitHub: <https://github.com/harshitsharma120109-web/ARIES-Search-Rescue-Rover> | Live Hub: <https://melodic-halva-b0fa56.netlify.app/>

'WE DON'T JUST BUILD ROBOTS, WE BUILD HOPE.' — Team ALPHA-X (SIH_1186)